



Administrativa

22 April 2025

Prof. Dr. Sebastian Wild

Goals for Today

- ▶ give you some detail on **what** this module covers

⇒ so that you can decide whether to keep it

if it is an elective module for you

Advanced Algorithms

- ▶ inform you about **how** AdvAlg is run
- ▶ inform you about how AdvAlg is **assessed**

Welcome to CS 627 – Advanced Algorithms

- ▶ Dozent: Prof. Dr. Sebastian Wild

Mehrzweckgebäude, Raum 05 D 16

wild@informatik.uni-marburg.de

Tutor: Mohammed Omer omerm@students.uni-marburg.de

- ▶ Module website: www.wild-inter.net/teaching/advalg

→ your first address for any infos on CS566

- ▶ *Campuswire*: collaborative Q&A (more on this later)

also used for announcements

→ please register via link from the ILIAS announcement

<https://campuswire.com/p/G8A144D4D>

PIN 9631

- ▶ *Slido*: student response system for formative feedback → bring a smart device to class!

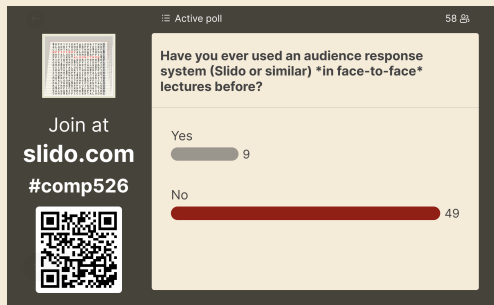
- ▶ Final mark: 100% final exam

Admission to exam: 50% of points from exercise sheets & mini-seminar

Audience Response System: *Slido*

- ▶ Goal: Collect immediate, formative feedback
 - ▶ Stay focused and engaged! (“active learning”)
 - ▶ Quick feedback (for you individually) if you are on track.
 - ▶ Quick feedback (for me) whether (most of) you are on track.
- ▶ Slido has 2 useful features:

1. Quick Polls



The screenshot shows the Slido interface for a quick poll. On the left, there is a QR code and the text "Join at slido.com #comp526". The main area displays the poll question: "Have you ever used an audience response system (Slido or similar) *in face-to-face* lectures before?". Below the question, there are two horizontal progress bars. The "Yes" bar is grey and shows 9 responses. The "No" bar is red and shows 49 responses. The top of the interface shows "Active poll" and "58" participants.

Join at
slido.com
#comp526

Have you ever used an audience response system (Slido or similar) *in face-to-face* lectures before?

Yes 9

No 49

2. Audience Questions



The screenshot shows the Slido interface for audience questions. On the left, there is a QR code and the text "Join at slido.com #comp526". The main area displays a list of questions under the heading "Q&A". The first question is from "Sebastian Wild" and asks "How can I ask a question in class?". The second question is from "Anonymous" and asks "I'm a bit unsure, I'd rather ask this anonymously.". The top of the interface shows "Q&A", "Popular", and "2" questions.

Join at
slido.com
#comp526

Q&A

Sebastian Wild
How can I ask a question in class?

Anonymous
I'm a bit unsure, I'd rather ask this anonymously.

My approach to lectures

My conclusions (from years of own experience, a pandemic, and observing others)

irrespective of the
mode of delivery!



- 0. Good explanations (intuitions!) and well-structure material are the most important aspect.
- 1. **Synchronous (live) lectures** beat videos in keeping up with class. (but recordings are great!)
- 2. Only a small minority of students asks questions in class. ~~~ other backchannels
- 3. **Interaction** makes content memorable (and keeps brains awake!) ~~~ *Slido* tasks

Components of EA

Slido questions

immediate feedback
simple questions

Lectures

new material
discussions
big picture

Tutorials

get practice solving problems
solve deep questions

Campuswire

collaborative Q&A knowledge base

Exam Question Gallery

collaborative pool of potential and past exam problems

Final Exam

summative assessment
of your acquired skills

Assessments

- ▶ **Module mark** = mark in final exam
- ▶ **Final exam**
 - ▶ written or oral examination
- ▶ **Exam Material:** everything covered in lectures (except marked “~~≠~~ exam”) everything covered in **tutorials** and exercise sheets
- ▶ **Admission requirements to final exam**
 - ▶ ≤ 2 exercise sheets with 0 points in your group (not handing in implies 0 points)
 - ▶ $\geq 50\%$ of available points in sum over all exercise sheets
 - ▶ successful participation in NP-completeness mini-seminar

stay tuned ...

NP-Completeness Mini-Seminar

► Goals

- Refresh NP-hardness and reductions
- Work with primary literature
- Practice writing and presenting formal proofs

► Deliverables

1. Written formal proof with full details
Must include two worked examples of the reduction (one Yes-instance, one No-instance)
2. Video presentation explaining your problem and the reduction
3. 3 short peer-review reports about other groups' submissions

► Organization

- During weeks 2 and 3
- Groups of 3–5 students
- On reductions from Richard Karp's original 21 NP-hard problems

Tutorials

▶ *Exercise Sheet* (Übungsblatt)

- ▶ released on module website
- ▶ to be **handed in**
 - ▶ handin on ILIAS
 - ▶ in **groups** of 3–4 students
- ▶ practice problems
- ▶ enhancement problems

▶ in *tutorials*

- ▶ discussion of selected solutions

▶ **Marking**

- ▶ Mainly for your feedback
- ▶ Serious attempts will yield partial credit even if unsuccessful

*Use the tutorials to **practice** your thinking!* = Don't cheat yourself!

*“If I tell you to run 10km,
it isn't because I want you
to be 10km away from me.”*

Generative AI

We live in exciting times!

LLMs (ChatGPT, DeepSeek etc.),
Media (Midjourney etc.), GitHub CoPilot, ...

- ▶ Generative Artificial Intelligence (GenAI) is amazing!
 - ▶ full of flaws (hallucination, bias, copyright, data privacy, cost, ...)
 - ▶ and yet ... often helpful, surprisingly versatile
- ▶ Why not use for everything?
 - ▶ Need for *deeply skilled* humans here to stay (for now anyways)
 - ↪ **Skill comes from practice!** (We still teach mental arithmetic in primary school!)



assessments designed for upskilling *humans*

↪ For our assessments:

*Don't take away the **thinking!** = Don't cheat yourself!*

Acceptable use:

- ▶ preparatory research (≈ Wikipedia)
- ▶ proof reading (spelling, grammar)

Unacceptable use: (not exhaustive!)

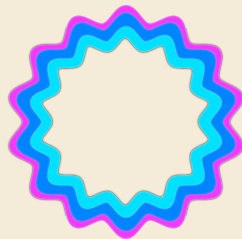
- ▶ use generated parts w/o acknowledgment & citation
- ▶ tools to paraphrase others' work to pass as own
- ▶ generated parts with inappropriate prompt, e. g., "write me a conclusion for this essay"

What is Campuswire?

Campuswire is an online space for lectures

1. ***Class Feed***: questions on material
2. ***Chatrooms***: structured social space
similar to Slack or Discord

We use Class Feed for **collaborative Q&A**



Join via link on website:
campuswire.com/p/G8A144D4D

Use in browser
campuswire.com/c/G8A144D4D

or via app
campuswire.com/download

- ▶ Ask ***public*** questions
 - ▶ “Why is $\lg(n^3) = \Theta(\log n)$?”
 - ▶ “Will there be classes on public holidays?”
- ▶ ***Answer* your peers’ questions!**
 - ▶ Know the answer? → put it in!
 - ▶ Know a partial answer? → Post it, others can build on it!
 - ▶ Found a helpful answer (or question)? → Vote it up!
- ▶ Ask ***private*** questions
 - ▶ if your question might contain “spoilers” for assessments
 - ▶ if you feel the answer is only relevant for you personally

How to Campuswire

- ▶ Our goals for Campuswire Q&A:

1. **be fair** Same answers for everyone
2. **learning by teaching** *You* should answer most questions (collectively)!
3. **be inclusive** posts can be anonymous; you can take your time to ask and answer

- ▶ Therefore, we instructors will

- ▶ redirect you to Class Feed for questions,
- ▶ wait before answering, to give other students a chance to answer first,
- ▶ explicitly mark good answers (and questions!) as such

ILIAS

- ▶ Official announcements
- ▶ Hand-in of exercise sheets
- ▶ Announcement of marks

... what can be on the public module website
goes to the public module website!



Exam Question Gallery

- ▶ We jointly collect a **pool of exemplary exam questions**.
- ▶ *You add **your** questions to it.*
- ▶ I will give feedback which questions are realistic.
- ▶ *... and we will pick one if there's sufficiently many good ones!*

↪ great resource for exam preparation

↪ We will answer selected questions in recap session (last week of classes)

- ▶ Engage in this early and pose great questions